

MATH3303 – Lecture 8

Automorphism Groups

Review Notes

Reference: Lee, Abstract Algebra, Section 4.6

Overview

This lecture introduces the automorphism group of a group and its most important subgroup, the inner automorphism group:

- **Definition** of an automorphism; $\text{Aut}(G)$ as a group under composition.
- **Examples:** $\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$; $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^*$; $\varphi : \text{GL}_n(\mathbb{R}) \rightarrow \text{GL}_n(\mathbb{R})$, $A \mapsto (A^T)^{-1}$ is an automorphism.
- **Inner automorphisms:** $\theta_a(g) = aga^{-1}$; the inner automorphism group $\text{Inn}(G)$.
- **Main theorems:** $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$; $G/Z(G) \cong \text{Inn}(G)$.
- **Outer automorphism group:** $\text{Out}(G) := \text{Aut}(G)/\text{Inn}(G)$.

1 The Automorphism Group

Definition 1.1: Automorphism; $\text{Aut}(G)$ (from lecture; cf. Lee, Def. 4.9)

Let G be a group. An *automorphism* of G is an isomorphism $\varphi : G \rightarrow G$.

The set of all automorphisms of G is called the *automorphism group* of G , denoted $\text{Aut}(G)$.

Remark 1.1: $\text{Aut}(G)$ is a group under composition (from lecture; cf. Lee, Thm. 4.21)

For any group G , $\text{Aut}(G)$ is a group under composition of functions:

- **Closure:** the composition of two isomorphisms is an isomorphism.
- **Associativity:** composition of functions is always associative.
- **Identity:** the identity function id_G is an automorphism.
- **Inverses:** every isomorphism has an inverse isomorphism.

Remark 1.2: Homomorphisms from cyclic groups are determined by the generator (from lecture)

In general, if G is a cyclic group generated by g , then a group homomorphism $\varphi : G \rightarrow H$ is completely determined by $\varphi(g)$.

In particular, if $G = \mathbb{Z}$ (generated by 1), then any group homomorphism $\varphi : \mathbb{Z} \rightarrow H$ is determined by $\varphi(1)$. For $n > 0$:

$$\varphi(n) = \varphi(\underbrace{1 + 1 + \cdots + 1}_n) = n\varphi(1),$$

and for $n < 0$, $\varphi(n) = -\varphi(-n) = -(-n)\varphi(1) = n\varphi(1)$.

2 Examples of Automorphism Groups

Example 2.1: $\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$ (from lecture; cf. Lee, Thm. 4.22(1))

We compute $\text{Aut}(\mathbb{Z})$, where $(\mathbb{Z}, +)$ is the additive group of integers.

Claim 1: any $\varphi \in \text{Aut}(\mathbb{Z})$ is determined by $\varphi(1)$.

By Remark 1.2, every homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ satisfies $\varphi(n) = n\varphi(1)$ for all $n \in \mathbb{Z}$.

Claim 2: $\varphi(1) \in \{\pm 1\}$ for any $\varphi \in \text{Aut}(\mathbb{Z})$.

Let $a = \varphi(1)$. Since φ is surjective, $\text{Im}(\varphi) = \{na : n \in \mathbb{Z}\} = \mathbb{Z}$. This forces $a = \pm 1$.

Conclusion. Therefore

$$\text{Aut}(\mathbb{Z}) = \{\varphi_1, \varphi_{-1}\},$$

where $\varphi_1(n) = n$ (the identity) and $\varphi_{-1}(n) = -n$. This is a group of order 2, hence

$$\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2.$$

Example 2.2: $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^*$ (exercise; cf. Lee, Thm. 4.22(2))

More generally, every automorphism $\varphi : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ is determined by $\varphi(1)$, and $\varphi(1)$ must be a generator of $\mathbb{Z}_n$, i.e. an element of $\mathbb{Z}_n^* = \{m \in \mathbb{Z}_n : \gcd(m, n) = 1\}$.

This yields an isomorphism $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^*$, where $\mathbb{Z}_n^*$ is a group under multiplication. (The bijection is $\varphi \leftrightarrow \varphi(1)$, where the group operation in $\text{Aut}(\mathbb{Z}_n)$ corresponds to multiplication in $\mathbb{Z}_n^*$.)

Example 2.3: $\varphi : \text{GL}_n(\mathbb{R}) \rightarrow \text{GL}_n(\mathbb{R}), A \mapsto (A^T)^{-1}$, is an automorphism (from lecture)

Define $\varphi : \text{GL}_n(\mathbb{R}) \rightarrow \text{GL}_n(\mathbb{R})$ by $\varphi(A) = (A^T)^{-1}$.

Group homomorphism.

$$\varphi(AB) = ((AB)^T)^{-1} = (B^T A^T)^{-1} = (A^T)^{-1} (B^T)^{-1} = \varphi(A) \varphi(B).$$

Injective. If $\varphi(A) = I_n$, then $(A^T)^{-1} = I_n$, so $A^T = I_n$, hence $A = I_n$. So $\ker \varphi = \{I_n\}$.

Surjective. For any $B \in \text{GL}_n(\mathbb{R})$, $\varphi((B^{-1})^T) = (((B^{-1})^T)^T)^{-1} = (B^{-1})^{-1} = B$.

Hence φ is an automorphism.

Remark. φ is *not* an inner automorphism of $\text{GL}_n(\mathbb{R})$ when $n \geq 2$. Suppose for contradiction that $\varphi = \theta_A$ for some $A \in \text{GL}_n(\mathbb{R})$. Then for all $B \in \text{GL}_n(\mathbb{R})$:

$$(B^T)^{-1} = ABA^{-1}.$$

Taking determinants: $\det((B^T)^{-1}) = \det(B)^{-1} = \det(ABA^{-1}) = \det(B)$.

This forces $(\det B)^2 = 1$, i.e. $\det B = \pm 1$ for all $B \in \text{GL}_n(\mathbb{R})$. But taking $B = 2I_n$ gives $\det(2I_n) = 2^n \neq \pm 1$ for $n \geq 1$. Contradiction.

(Exercise: for $\text{SL}_n(\mathbb{R})$, φ is inner when $n = 2$, but *not* when $n \geq 3$.)

3 Inner Automorphisms

Definition 3.1: Inner automorphism; $\text{Inn}(G)$ (from lecture; cf. Lee, Def. 4.10)

Let G be a group and $a \in G$. The *inner automorphism induced by a* is

$$\theta_a : G \rightarrow G, \quad \theta_a(g) = aga^{-1}.$$

The *inner automorphism group* of G is

$$\text{Inn}(G) := \{\theta_a : a \in G\}.$$

Remark 3.1: Note on conjugation convention (from lecture)

Some texts (including Lee, Def. 4.10) define $\theta_a(g) = a^{-1}ga$. The lecture uses $\theta_a(g) = aga^{-1}$. The two conventions produce isomorphic groups; the key properties below hold in either case.

Lemma 3.1: $\theta_a \in \text{Aut}(G)$ for every $a \in G$ (from lecture; cf. Lee, Lemma 4.2)

For any $a \in G$, θ_a is an automorphism of G .

Proof. Homomorphism. For $g, h \in G$:

$$\theta_a(gh) = a(gh)a^{-1} = (aga^{-1})(aha^{-1}) = \theta_a(g)\theta_a(h).$$

Injective. Suppose $\theta_a(g) = 1_G$, i.e. $aga^{-1} = 1_G$. Then

$$ag = a \implies g = 1_G \quad (\text{cancellation law}).$$

So $\ker \theta_a = \{1_G\}$, and θ_a is injective.

Surjective. For any $g \in G$: $\theta_a(a^{-1}ga) = a(a^{-1}ga)a^{-1} = g$.

Hence $\theta_a \in \text{Aut}(G)$. □

Lemma 3.2: Composition law and inverses for inner automorphisms (from lecture; cf. Lee, Lemma 4.2)

For any $a, b \in G$:

- (1) $\theta_a \circ \theta_b = \theta_{ab}$;
- (2) $(\theta_a)^{-1} = \theta_{a^{-1}}$.

Proof. (1) For any $g \in G$:

$$(\theta_a \circ \theta_b)(g) = \theta_a(\theta_b(g)) = \theta_a(bgb^{-1}) = a(bgb^{-1})a^{-1} = (ab)g(ab)^{-1} = \theta_{ab}(g).$$

(2) By (1), $\theta_a \circ \theta_{a^{-1}} = \theta_{aa^{-1}} = \theta_{1_G} = \text{id}_G$, and similarly $\theta_{a^{-1}} \circ \theta_a = \text{id}_G$. □

4 $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$

Theorem 4.1: $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$ (from lecture; cf. Lee, Thm. 4.23)

Let G be a group. Then $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

Proof. (1) $\text{Inn}(G) \leq \text{Aut}(G)$.

- Each $\theta_a \in \text{Aut}(G)$ by Lemma 3.1, so $\text{Inn}(G) \subseteq \text{Aut}(G)$.
- **Closed under composition:** $\theta_a \circ \theta_b = \theta_{ab} \in \text{Inn}(G)$ by Lemma 3.2(1).
- **Closed under inverses:** $(\theta_a)^{-1} = \theta_{a^{-1}} \in \text{Inn}(G)$ by Lemma 3.2(2).
- **Identity:** $\theta_{1_G} = \text{id}_G \in \text{Inn}(G)$.

So $\text{Inn}(G) \leq \text{Aut}(G)$.

(2) **Normality:** $\alpha \circ \theta_a \circ \alpha^{-1} \in \text{Inn}(G)$ for all $\alpha \in \text{Aut}(G)$, $\theta_a \in \text{Inn}(G)$.

For any $g \in G$:

$$\begin{aligned} (\alpha \circ \theta_a \circ \alpha^{-1})(g) &= \alpha(\theta_a(\alpha^{-1}(g))) \\ &= \alpha(a \cdot \alpha^{-1}(g) \cdot a^{-1}) \\ &= \alpha(a) \cdot \alpha(\alpha^{-1}(g)) \cdot \alpha(a^{-1}) \\ &= \alpha(a) \cdot g \cdot (\alpha(a))^{-1} \\ &= \theta_{\alpha(a)}(g). \end{aligned}$$

Hence $\alpha \circ \theta_a \circ \alpha^{-1} = \theta_{\alpha(a)} \in \text{Inn}(G)$.

Therefore $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$. □

5 $G/Z(G) \cong \text{Inn}(G)$

Theorem 5.1: $G/Z(G) \cong \text{Inn}(G)$ (from lecture; cf. Lee, Thm. 4.24(2))

Let G be a group. Then

$$G/Z(G) \cong \text{Inn}(G).$$

Proof. Define

$$\varphi : G \longrightarrow \text{Inn}(G), \quad \varphi(a) = \theta_a.$$

φ is a group homomorphism. For $a, b \in G$:

$$\varphi(ab) = \theta_{ab} = \theta_a \circ \theta_b = \varphi(a)\varphi(b),$$

using Lemma 3.2(1).

φ is **surjective** by definition: every element of $\text{Inn}(G)$ has the form $\theta_a = \varphi(a)$.

$$\ker \varphi = Z(G).$$

$$a \in \ker \varphi \iff \theta_a = \text{id}_G \iff aga^{-1} = g \text{ for all } g \in G \iff ag = ga \text{ for all } g \in G \iff a \in Z(G).$$

Apply the First Isomorphism Theorem (for groups).

By the First Isomorphism Theorem,

$$G/\ker \varphi \cong \text{Im}(\varphi),$$

i.e.

$$G/Z(G) \cong \text{Inn}(G).$$

□

Remark 5.1: Consequence: $\text{Inn}(G)$ is trivial iff G is abelian (from lecture)

Since $G/Z(G) \cong \text{Inn}(G)$:

- If G is abelian, then $Z(G) = G$, so $\text{Inn}(G) \cong G/G = \{1\}$.
- Conversely, if $\text{Inn}(G) = \{\text{id}_G\}$, then $Z(G) = G$, so G is abelian.

Inner automorphisms are only interesting in *nonabelian* groups.

6 The Outer Automorphism Group

Definition 6.1: Outer automorphism group (from lecture)

The *outer automorphism group* of G is the quotient

$$\text{Out}(G) := \text{Aut}(G) / \text{Inn}(G).$$

This is well-defined since $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ (Theorem 4.1).

Remark 6.1: Terminology (from lecture)

Elements of $\text{Out}(G)$ are cosets $\alpha \cdot \text{Inn}(G)$. An automorphism $\alpha \in \text{Aut}(G)$ is called an *outer automorphism* if $\alpha \notin \text{Inn}(G)$.

Note: the “outer automorphism group” is not the group of outer automorphisms — it is the *quotient* group. Individual outer automorphisms are elements of $\text{Aut}(G)$, not of $\text{Out}(G)$.

Example 3.1 shows that $A \mapsto (A^T)^{-1}$ is an outer automorphism of $\text{GL}_n(\mathbb{R})$ for $n \geq 2$.

Checklist of Skills

- ☐ State the definition of an automorphism and the automorphism group $\text{Aut}(G)$.
- ☐ Explain why $\text{Aut}(G)$ is a group under composition.
- ☐ Show that any homomorphism from a cyclic group is determined by the image of a generator.
- ☐ Compute $\text{Aut}(\mathbb{Z})$ and deduce $\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$.
- ☐ State (and verify) the isomorphism $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^*$.
- ☐ Verify that $A \mapsto (A^T)^{-1}$ is an automorphism of $\text{GL}_n(\mathbb{R})$.
- ☐ Show it is not inner using a determinant argument.
- ☐ Define the inner automorphism $\theta_a(g) = aga^{-1}$ and the group $\text{Inn}(G)$.
- ☐ Prove $\theta_a \in \text{Aut}(G)$ (homomorphism, injective, surjective).
- ☐ Prove the composition law $\theta_a \circ \theta_b = \theta_{ab}$ and the inverse law $(\theta_a)^{-1} = \theta_{a^{-1}}$.
- ☐ Prove $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$: subgroup (closure, inverses, identity) and normality ($\alpha\theta_a\alpha^{-1} = \theta_{\alpha(a)}$).
- ☐ Prove $G/Z(G) \cong \text{Inn}(G)$ via the map $\varphi(a) = \theta_a$ and the First Isomorphism Theorem.
- ☐ Identify $\ker \varphi = Z(G)$ in the above proof.
- ☐ Explain why $\text{Inn}(G)$ is trivial iff G is abelian.
- ☐ Define the outer automorphism group $\text{Out}(G) = \text{Aut}(G)/\text{Inn}(G)$.